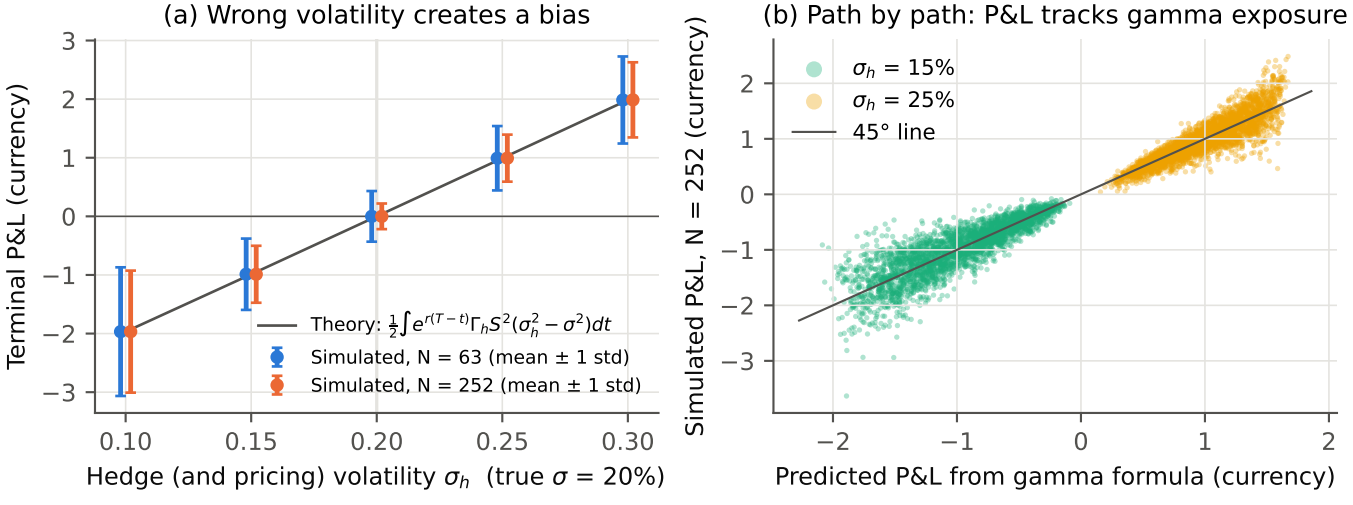

(a) Wrong volatility creates a bias
Terminal P&L (currency)
Hedge (and pricing) volatility σ_h (true σ = 20%)
Theory: $\frac{1}{2}\int e^{r(T-t)}\Gamma_h S^2(\sigma_h^2-\sigma^2)dt$
Simulated, N = 63 (mean ± 1 std)
Simulated, N = 252 (mean ± 1 std)
(b) Path by path: P&L tracks gamma exposure
Simulated P&L, N = 252 (currency)
Predicted P&L from gamma formula (currency)
σ_h = 15%
σ_h = 25%
45° line